International Journal of Rock Mechanics & Mining Sciences 147 (2021) 104912

Contents lists available at ScienceDirect

International Journal of Rock Mechanics and Mining Sciences

journal homepage: www.elsevier.com/locate/ijrmms

Technical Note

Using acoustic impacts and machine learning for safety classification of mine roofs

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ARTICLE INFO

Keywords:
Safety
Mine roof stability
Machine learning
Classification
Acoustics

ABSTRACT

The safety of mine roofs is often evaluated by listening to the sound of a scaling bar impact: well-supported ground sounds "tight" while potentially unsafe ground is described as "drummy". This is a subjective process that requires training and experience before a mine worker can confidently assess the sound. This paper presents a method of removing subjectivity from this process. The authors recorded over 3000 impacts from a variety of potash mines, which were labeled as "tight" or "drummy" by experienced workers. We then used machine learning techniques to classify these recordings and were able to achieve classification errors of less than 1% on new data not used to train the algorithms. Alternately, we could bring false-negative results to zero with a small increase in false-positives. This paper also presents a number of different classification and data dimensionality reduction techniques and compares their effectiveness on this dataset.

1. Introduction

Mine roof stability is of critical importance to underground miners. Falls of roof material can cause injury or death to workers, or equipment damage or loss. One common method of evaluating the safety of a roof in using scaling bar impacts.[1] Before entering a newly mined area or before starting work in an area, the worker will use a scaling bar (a steel or aluminum bar usually 2–3 m in length) to strike the roof in various locations. By listening to the sound of the impact, the worker can estimate whether that area of roof is likely to fall.

In the present study, the authors were concerned with Saskatchewan potash mines. These mines are characterized by sedimentary layers of relatively strong potash ore (primarily KCl and NaCl salts), alternating with weaker clay seams. The location of these clay seams relative to mine opening are usually well understood, and the mine is designed so that a thick layer of salt is left above the opening to support the roof. When struck with a scaling bar this strong, massive layer damps out the vibrations caused by the impact and the sound return is said to be "tight". (These somewhat imprecise terms are used in this paper as they are well understood by local miners). However, if a weak clay layer dips down toward the opening, the thin salt layer is weakened and can separate from overlying layers. This presents a dangerous situation, prone to rock falls. In this case, the salt beam can experience more resonant vibrations and the sound is termed hollow-sounding or "drummy". Upon detecting drummy ground, the workers immediately retreat from the area and install expanding shell or epoxied roof bolts, in order to attach the thin salt layer to overlaying strong layers. Currently roof bolts are installed when drummy ground is observed or when mining procedures require installation, however automatic installation of roof bolts while mining is being investigated to improve operational efficiency.[2]

Evaluating whether an impact sounds tight or drummy is a subjective process that requires training and experience. Although no fatalities have been attributed to misclassification of these impacts, unconfidence and an understandable bias toward caution in novice workers can cause the roof bolting procedure to be falsely triggered. Additionally, some conditions can be difficult even for experienced miners, such as intersecting rooms and high roofs. This paper aims to develop a method of automatically classifying mine roof impacts, removing the subjective human factor from the process. As mines move toward automated machinery, this method may also allow for remote evaluation of roof condition without a human entering the mine.

Some of the previous research has used impacts to evaluate mine safety, but it largely focuses on local rock properties.[3,4] In this method, a Schmidt Hammer uses the elastic rebound of an impacting mass to estimate the compressive strength of rock. Although this hammer was originally designed for concrete testing it was applied to evaluating rock properties. This method proved to be time consuming and did not perform well as many repetition were required. A patent by Choi[5] presents an acoustic method of evaluating mine roof bolts, based the their natural frequency. Unfortunately the mine roof itself is much more damped and has a complex frequency response with no obvious

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https://doi.org/10.1016/j.ijrmms.2021.104912
Received 8 September 2020; Received in revised form 25 August 2021; Accepted 9 September 2021
Available online 7 October 2021
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natural frequency. In another patent, Gallagher[1] presented a method in which the roof is struck with a hammer and a geophone drilled in at a known location senses vibrations which may be used to interpret discontinuities in the rock. On a much larger scale, seismic exploration techniques can be applied, where an impact or wide-band vibrational signal is used to excite the ground and the reflection time is used to identify the change in the structure of the rock. The exciter can be a hammer, weight-drop or hydraulic vibrator. A conceptually similar technology is seen in the oil and gas industry where the ultrasonic inspection of cemented well casings is used.[6] Wave propagation of electromagnetic waves have also been used to characterize the structure of mine roofs, including Ground Penetrating Radar.[7] All the methods described above are both costly and time consuming methods, while the scaling bar impact test is fast and low-cost.

Although the literature in the area of acoustic impact classification and mining is sparse, there is considerable research on machine learning for acoustics. Some examples include speech recognition,[8] sound localization,[9] reverberation removal,[10] and nondestructive testing of materials.[11] A recent review can be found in[12].

This paper first presents our methodology for collecting a dataset of impacts from mine sites. A description of the classification and dimensionality reduction algorithms is then presented. Results are then shared, showing the comparative performance of the various algorithms implemented.

2. Dataset collection

In order to test and compare classification schemes, a library of labeled roof impacts was assembled. We made 6 visits to a variety of potash mine sites in Saskatchewan. At each site, a working day was spent traveling through the mine to various locations with varying roof conditions. As shown in Fig. 1, for each recording a Sennheiser E 845-S microphone was mounted on a tall microphone stand such that the microphone was located near to the impact site (previously, preliminary data was recorded at ear-height, but it was determined that the room acoustics provided too much interference). A Zoom H4n Pro portable audio recorder was then activated to record a 24-bit wav file at a rate of 48 kHz. The roof was then struck with a scaling bar, with approximately the same strength of impact that would normally be used. The mine worker would then label the impact as “tight” or “drummy”. Safety concerns prevented obtaining recordings of unsupported drummy ground so drummy recordings were taken from areas that had roof bolts installed. It was reported that the sound of drummy ground does not significantly change after roof bolts are installed. We recorded a total of 3309 labeled impacts, each 2–4 s long. Further details, along with the data files, are available from[13].

Some preprocessing was performed on the dataset. Each impact was identified and aligned in time by identifying the first sample with a magnitude exceeding ten times the noise floor. The impact was then cropped to 36 000 samples (0.75 s) with 100 samples preceding the trigger. In order to compensate for the varying impact strength, the data was normalized to unity variance. The first M samples of the N recordings were then assembled into a N x M data matrix, $\boldsymbol{X} \in R$. The data labels were then used to assemble a N x 1 response vector, $\mathbf{Y}$ where 0 represents a recording labeled as drummy and 1 represents “tight”. A sample of the data is shown in Figs. 2 and 3. On initial inspection, the difference between the drummy and tight recordings is not immediately obvious in either the time or frequency domain data. It was believed that the difference between tight and drummy sounds would be more noticeable in the frequency domain, so the fast Fourier transform was applied to the data, discarding the $\omega = 0$ mean value and keeping the absolute value of $M/2-1$ remaining values below the Nyquist frequency. Other combinations of predictors were evaluated, as discussed in Section 4.3.

Although collecting 3309 impacts represented a considerable investment of time, considering the number of data points in each recording, this represents a classification problem with more predictors than observations, so overfitting is a concern. The dataset is also quite noisy, including external and internal noise sources (although external acoustic noise sources, such as wind and machinery dominate). Lastly, the dataset is characterized by high correlation between predictors (correlation matrices are shown in Fig. 4 for the first $M = 6000$ points), indicating that the dataset is highly redundant and the effective dimensionality is smaller than the number of predictors.

Fig. 1. Typical experimental data collection setup, including worker striking roof with scaling bar, and microphone and digital audio recorder mounted on tripod, with microphone located near impact site.

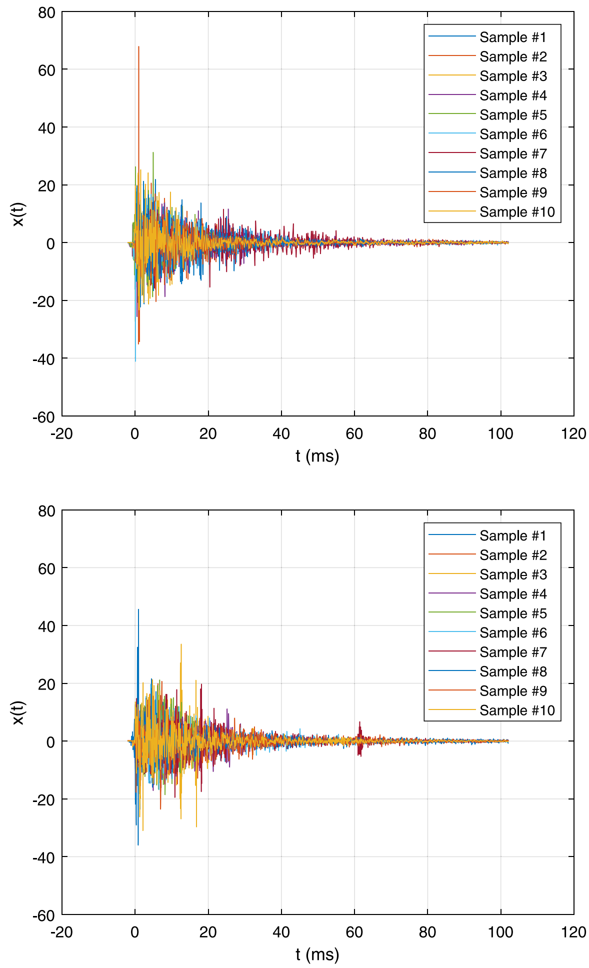

Fig. 2. Time domain results for 10 sample impacts from mine visit 1, after normalization to unity variance, for impacts labeled "tight" (top) and "drummy" (bottom). There is no obvious feature separating the two classes of response.

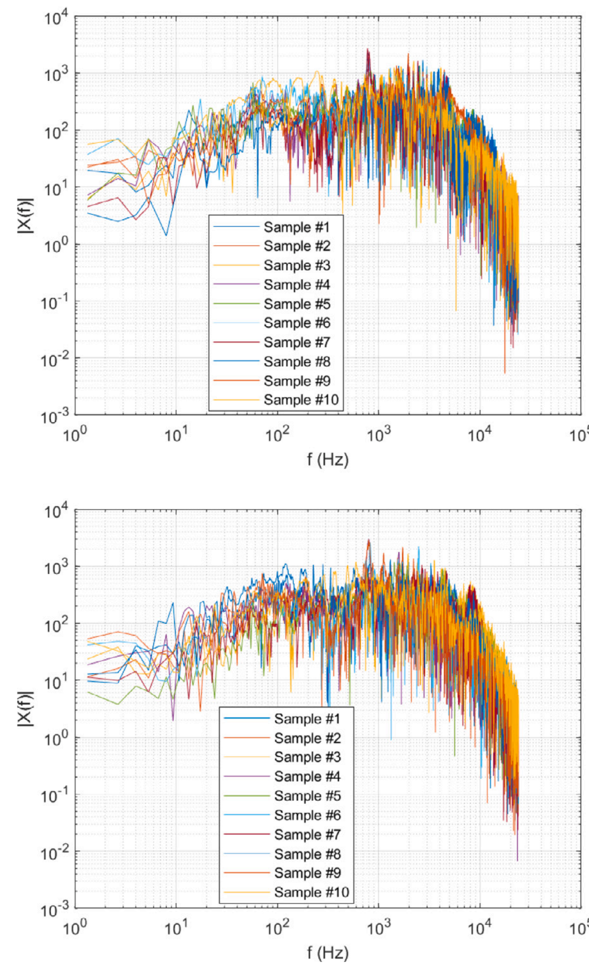

Fig. 3. Frequency domain results for 10 sample impacts from mine visit 1, after normalization to unity variance, for impacts labeled "tight" (top) and "drummy" (bottom). Like Fig. 2, there is no immediately obvious difference.

3. Classification

The goal is to predict whether a previously unseen recording would be labeled as "drummy" (0) or "tight" (1) by an expert worker. While there is a continuum of "drumminess", we have treated this as a binary classification based on whether the roof would be considered drummy enough to trigger ground support procedures. This classification is based on a subjective "ground truth" (i.e. an expert worker's opinion rather than measurements), but this can be justified by the fact that these experts have a long track record of successfully identifying unsafe conditions, which are later verified by methods such as drilled cores or ground penetrating radar.

The prediction process has two portions: feature generation and selection, and classification. Each impact's audio recording has 36 000 data points, but they are highly correlated and some are irrelevant, so it is believed that a much smaller set of features can be used to more concisely describe each impact. By reducing the number of features, the "curse of dimensionality" is reduced, allowing for the classification task to be performed more easily and with better ability to extrapolate to recordings taken under different conditions (avoiding overfitting). One method would be to extract features manually using physical parameters, such as decay time, peak power, etc. However, this requires considerable knowledge of the physics of the problem and it was unknown if enough conventional physical features existed to describe the impacts with sufficient accuracy.

Instead, a feature reduction method called Principal Component Analysis (PCA) was used.[14] This is a linear technique which uses an eigen-analysis of the covariance of the data to recompose the data into principal components, each with minimal correlation to the others, and sorted by the magnitude of their contribution to the variance (equal to the eigenvalues, λ). These principal components do not have a physical meaning, but one can select a small number of them to represent the data (with small loss in accuracy but increased generalization).

A number of methods exist for selecting which principal components are used for the classification.[15] These include scoring the features by

- explained variance, λ
- correlation with the response vector
- probability given by a two-sample t-test between the feature and the response vector[16,17]
- mutual information[18]

Once the J best features were selected, a classification algorithms was applied. The authors investigated a number of existing algorithms,[19] including

- linear regression
- logistic regression
- linear discriminant analysis (LDA)
- linear and quadratic support vector machines
- naive Bayes classification

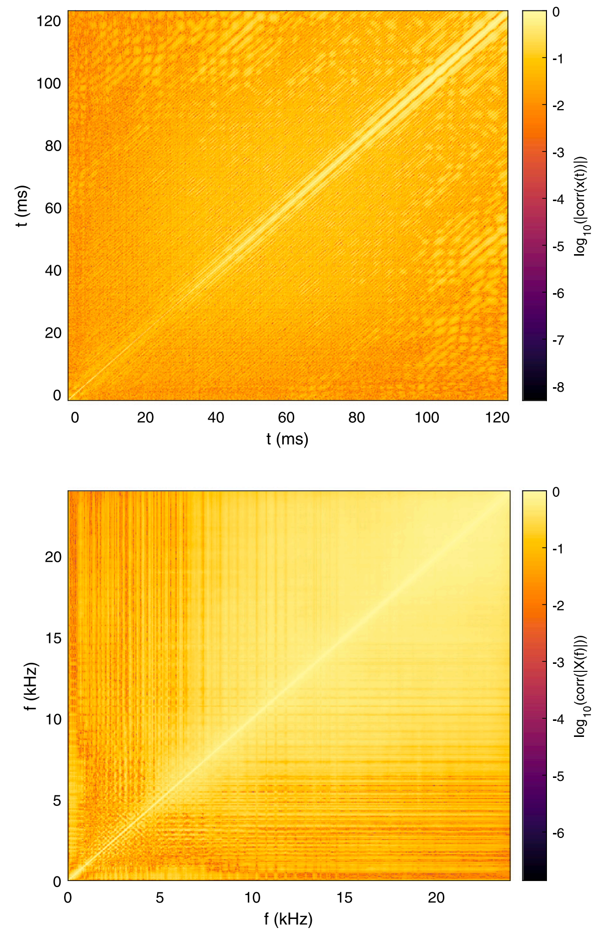

Fig. 4. Correlation matrices between predictors for time domain data (top) and frequency domain (bottom). The diagonal banding structure in the time domain data is likely due to electrical noise at 60 Hz.

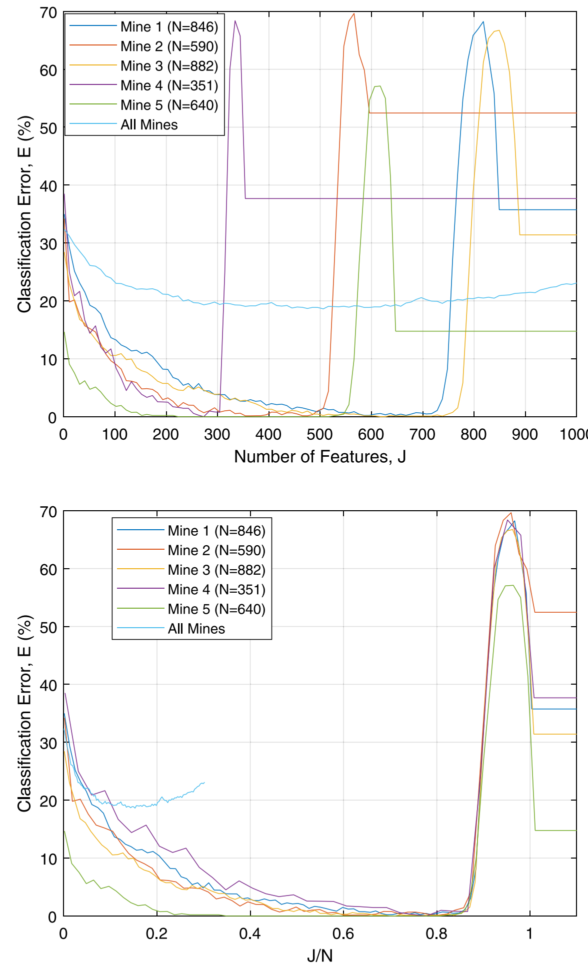

Fig. 5. Classification error for each of the 5 mine visits, along with the combined data set from all mines together. The same data is shown at right, normalized by the number of observations, N.

With exception of the quadratic support vector machine, all methods assume linear decision boundaries between classes. These algorithms were selected as they are well known in the literature and provide a good baseline for performance. Other more sophisticated algorithms will be investigated in future work, but are not evaluated here.

To evaluate the performance of each method, K-fold validation was used.[20] In this validation scheme, the data set is randomly divided into K subsets of the data ($K = 10$ here). One subset is reserved for validation and the remaining data is used to generate the regression model. The validation data is then used to evaluate the accuracy of the model using the fraction of the data properly classified. This is then repeated K times for the different validation subsets and the errors averaged. This ensures that the classifier is being evaluated on data it has not seen before and reduces the effect of random biases.

4. Results

4.1. Number of features

One of the first questions to be explored was how many features (principal components) were required to accurately classify the data. Using LDA , a data length $M = 3000$, and each mine visit treated separately, the K-fold validation classification error was calculated while the number of features selected (sorted by correlation) varied up to 1000, with results shown in Fig. 5. For each mine's dataset, the classification error strongly reduces with J until it reaches a minimum using around $J = 0.8N$ principal components, with classification errors below 1.0%. This is considerably fewer than the 3000 points in each impact recording, demonstrating the ability to reduce the dimensionality of the problem while maintaining the ability to classify the data accurately. The analysis was also repeated combining all five datasets into one large dataset. As shown in Fig. 5, the classification error in this case is not reduced below 18%. This indicates that a single linear model cannot be used to represent the conditions in the different mines and caution should be used when extrapolating.

4.2. Data length

In the preceding analysis each impact was trimmed to $M = 3000$ points (62.5 ms of data). How the value of M influenced the classification error was investigated next. As shown in Fig. 6, the minimum error decreases with M until it levels out around $M = 1900$ (40 ms), indicating that the critical information for classifying the impacts can be found in these initial stages of the recording. This is favorable, as it indicates that the hardware required to implement this algorithm will not be required to record long impact responses, and memory requirements will be modest.

4.3. Preprocessing

The effect of input preprocessing was also investigated. The above analysis uses the magnitude of the Fourier transform of the sound waveform as a the input. As shown in Fig. 7, this was repeated adding

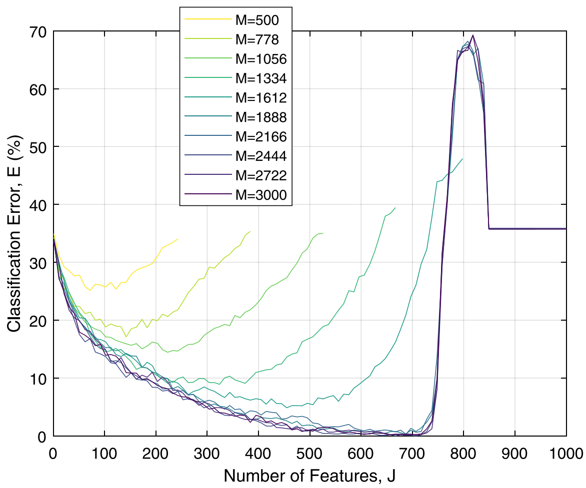

Fig. 6. Classification error while varying M, the length of data used for each impact. This shows results for mine visit 1, but other visits showed similar trends.

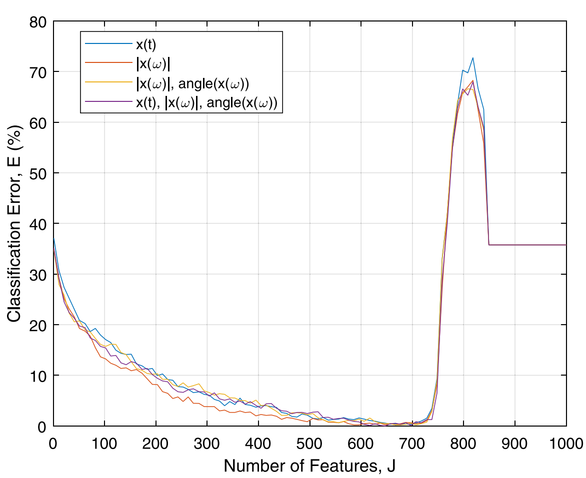

Fig. 7. Effect of selecting different inputs for the classifier, including the impact sound's time series, and/or frequency domain magnitude and phase.

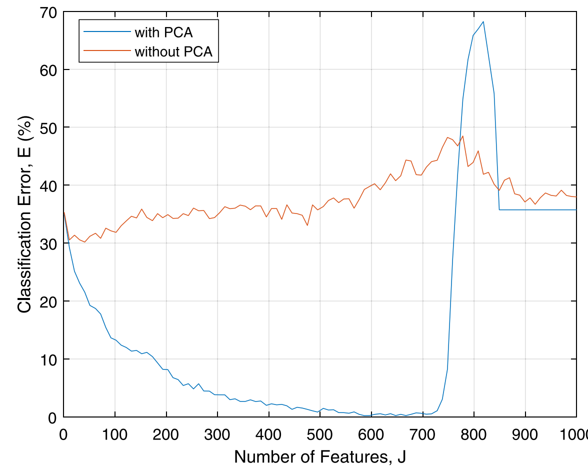

Fig. 8. Effect of principal component analysis.

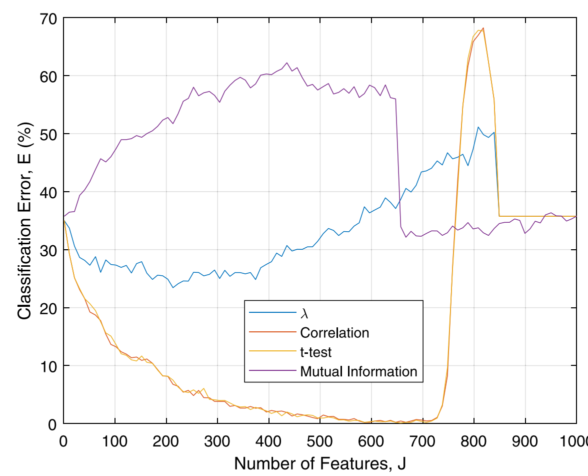

Fig. 9. Classification error for mine visit 1 and M=3000, while changing the feature selection algorithm. Similar results were obtained from other mine visits.

the time-domain input, and phase angle of the frequency-domain data. The data using only the Fourier magnitude provides the best classification, although the difference between methods becomes insignificant as the number of features increases. Thus, if computation power is at a premium, the raw time-domain data can be used, avoiding the Fourier transform calculation.

We also investigated the effectiveness of the principal component analysis by comparing it to a classifier in which the raw data is used, with no PCA performed. As shown in Fig. 8, without the PCA, the classification error is comparable to a constant model (a constant model would have $E = 36\%$); clearly the dimensionality reduction of the PCA reduces the effect of the redundant nature of the data.

4.4. Feature selection and classification algorithms

Next, the effect of the feature selection and classification algorithms were investigated. As described above, the principal components used in the regression can be selected by a number of methods, each with different assumptions and advantages and weaknesses. Each method was applied using LDA classification. As shown in Fig. 9, the correlation and t-test methods shows considerably lower classification error than the λ method across the range of principal components, indicating that a considerable number of significant principal components are not well correlated and can be eliminated. The mutual information method exhibits poor performance, likely because, as a kernel method, there is not enough information to generate an accurate estimate of the probability functions from the data.

The effect of the classification scheme was also examined, with results in Fig. 10. Generally, the linear discriminant analysis and linear regression analysis showing the smallest error rates, with logistic regression close behind. The difference between these three schemes becomes insignificant near the minima. The linear and quadratic support vector machines show slightly poorer performance and exhibit a higher minimal error. For comparison, the naive Bayesian classifier exhibits very poor performance (close to a constant model), which can be expected as it assumes no correlation between predictors, whereas this dataset is highly correlated.

Based on this, one can conclude that a wide range of traditional classification algorithms can be applied successfully, but that the designer's effort and consideration should be concentrated on feature selection algorithms and training dataset composition.

4.5. Cross-mine validation

One of the objective of this project is to understand whether a model made from one mine data can be applied in another mine. In this section the classification model is developed using one mine's data and tested with the data obtained from other mine.

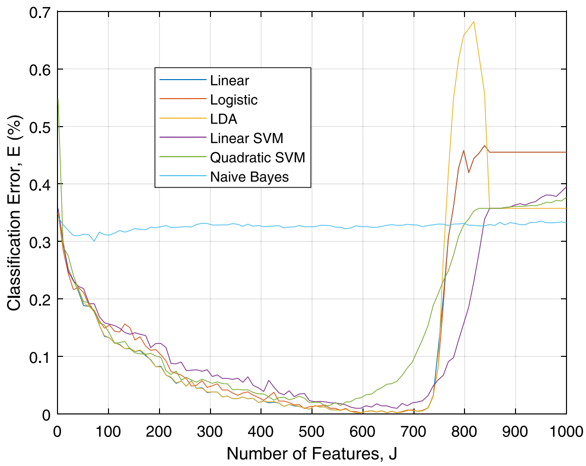

Fig. 10. Classification error for mine visit 1 and M=3000, while changing the classification algorithm. Similar results were obtained from other mine visits.

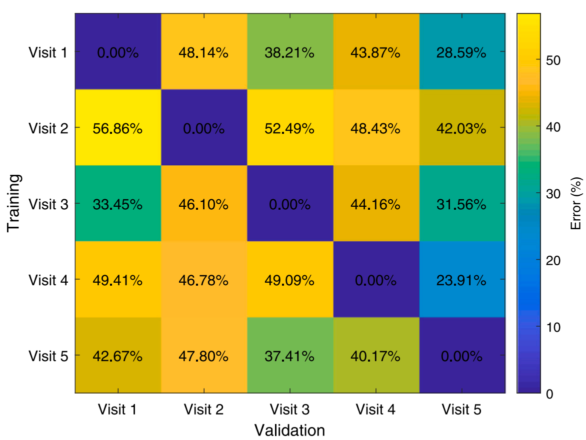

Fig. 11. Cross-mine validation, with the classifier trained using data from one mine visit and evaluated on another.

As we had previously observed that the algorithm performs well for $M = 3000$ points and for $J = 0.8N$ features, these parameters were used to train a model using one mine's data and evaluated using a different data set. Fig. 11 shows that all the models of the mines produce very low error for testing data on themselves (note this does not use K-fold validation, so this is not predicting new data and will overestimate performance for same-mine conditions), however for cross-mine validation we see error as high as 57% and as low as 23%. This result clearly indicates that mine models should not be applied to other mines unless additional precautions are taken.

4.6. Threshold effects

Another feature of this type of algorithm is the ability to apply a controlled bias to the classification of marginal data. For example, safety dictates that one should prefer that a safe situation be erroneously classified as unsafe, rather than vice versa. This can be achieved by modifying the decision rules for a classifier. For example, support vector machines can report the estimated posterior probability of a measurement falling into a class. Rather than selecting the class with the highest probability, one can specify a threshold probability that must be met in order to consider the observation to be part of a certain class. For example, Fig. 12 shows how the false positive and negative rates are affected by this threshold probability. One can then adjust this bias to reduce the false negative rate (i.e. identifying an

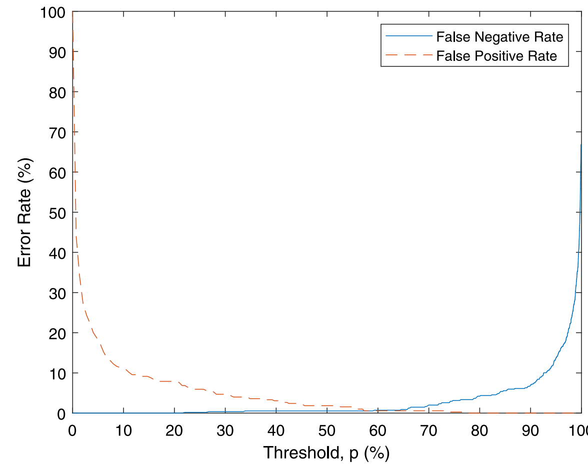

Fig. 12. Binary classification with a varying threshold probability of being classified as "drummy". This was calculated using a linear support vector machine for visit #1 data, $M = 3000$, $J = 0.8N$, and features selected by correlation.

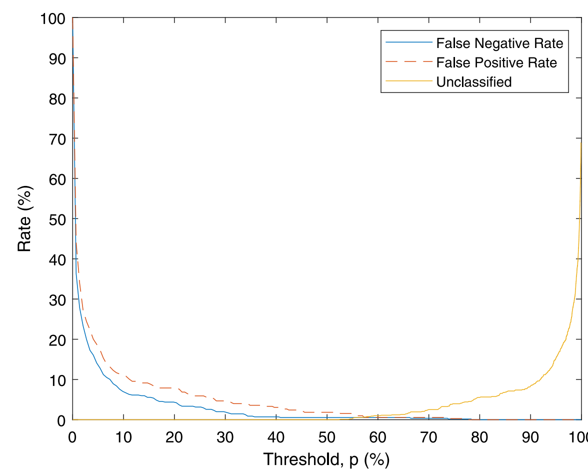

Fig. 13. Classification including an "unknown" option for classifications with a posterior probability below the threshold p. This was calculated using a linear support vector machine for visit #1 data, $M = 3000$, $J = 0.8N$, and features selected by correlation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

observation as tight when it is actually drummy), at the cost of a slightly increased false positive rate.

Another option for increased safety is to add a third output option for the classifier. Rather than deciding if the observation is one of "drummy" or "tight", one can add the option of "unknown". This can be implemented by requiring a certain posterior probability in order to classify as either "drummy" or "tight", and if it does not reach this probability for either, this third "unknown" result is returned. Fig. 13 shows the resulting error curve and the fraction of observations that would be classified as unknown. This allows the false negative rate to be reduced to zero on our dataset at the cost of 5% unclassified observations. In a device, this could be displayed to the user as a green light for "tight", yellow for "unknown", or red for "drummy".

5. Conclusion

While preliminary, this paper has shown the feasibility of using machine learning algorithms to remove the subjectivity from the process of evaluating mine roof stability using scaling bar impacts. This has shown that an important part of this process is feature generation using principal component analysis and feature selection using either correlation or a t-test. The number of features included in the classification has a strong effect on the classification accuracy with the best performance occurring when setting the number of features around 80% of the number of observations. The results are not particularly sensitive to the classification algorithm, with a number of algorithms giving comparable error rates. Similarly, although applying the Fourier transform to the data made a small improvement on the accuracy, the algorithm is not strongly sensitive to different inputs being included. This paper also demonstrates the ability to handle marginal cases, either by biasing the classifier away from false negative errors, or by adding an "unknown" output if the posterior probability is too low.

One important observation was that a classifier trained on data from one mine was not able to make accurate predictions on data from another mine. It is not yet clear how likely this effect would be to appear if a new section of mine was opened which had differing geology. Also, these results show feasibility for the potash mines tested on, but it is not clear how well this will apply to other mines, especially hard rock mines where the auditory difference between roof conditions is less pronounced. Other directions for future work will include the effect of noise and other confounding effects, as well as more sophisticated classification algorithms.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was funded by the International Minerals Innovation Institute, including funding organizations Mosaic Potash and Nutrien. The authors would like to acknowledge the assistance of mine engineers and technicians Andrew Derkewych, Edward Kowalyk, Nathan Morgan and Melanie Stare in creating and labeling recordings.

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